

$$\lambda = hc / (E_3 - E_1) = hc / (1303 - 49.7)(1.6 \times 10^{-19}) = 0.992 \text{ nm}$$

30.

Ans: A

Intensity

= Power / Area

= Energy / (Time)(Area)

= Nhf/tA

= $Nhc / \lambda tA$

→ λ is inversely proportional to intensity.

Since intensity is halved, then λ is doubled. Using $\lambda = h/p$, momentum of photon is halved when λ is doubled.